CS 3244 Machine Learning

Week 13 – Ethics

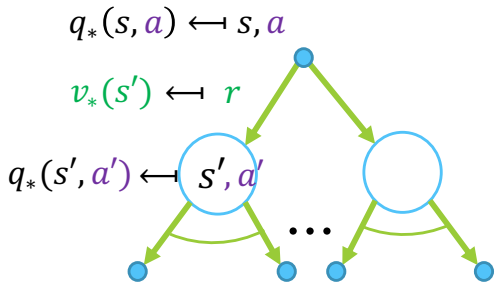
Kan Min-Yen

13 pre

Background Photo credits: Rafiq Mirza, Luan Ahn, Rosmarie Voegtli @ Flickr

Recap and Forecast

- Understand the basic anatomy and terminology in reinforcement learning
  - Policy, reward, action, observation and (environment, agent) model
- Introduce the Markov property and use this to define a Markov decision process (MDP)
- Understand the Bellman equation as a recursive form for calculating future rewards.



$$q_*(s,a) = \mathcal{R}_s^a + \gamma \sum_{s' \in \mathcal{S}} \mathcal{P}_{s,s'}^a \max_{a'} q_*(s',a')$$

Learning Outcomes for this week:

- Let's get FAT (i.e., Fairness, Accountability and Transparency)
- Have awareness of ML in the larger context, especially how it can affect people's lives other than ourselves;
- Understand the systematic shortcomings of problems with machine learning, in its stages of:
  - Data gathering;
  - Preparation;
  - Model learning.

ML Gone Wrong

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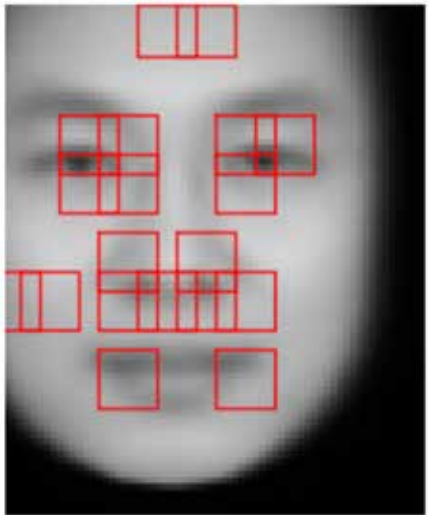
Credits: Tech Republic. *Why Microsoft's Tay AI Bot Went Wrong.* Source: https://www.techrepublic.com/article/why-microsofts-tay-ai-bot-went-wrong/

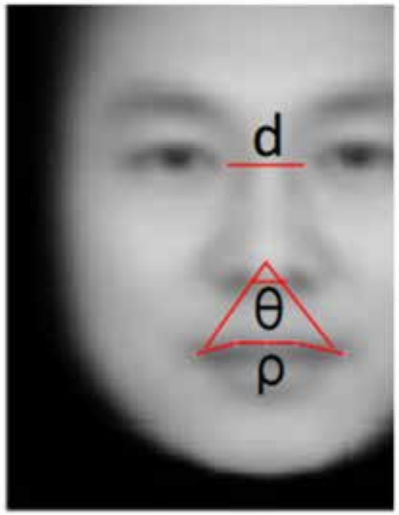
Connectivity

Neural Network Learns to Identify Criminals by Their Faces

The effort aimed at identifying criminals from their mugshots raises serious ethical issues about how we should use artificial intelligence.

by Emerging Technology from the arXiv November 22, 2016

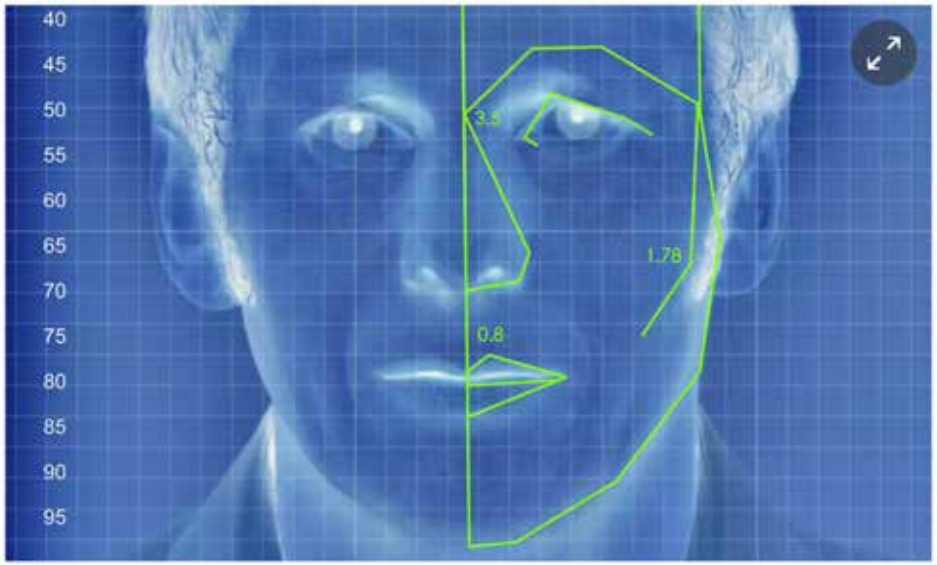




Credits: MIT Technology Review. Source: https://www.technologyreview.com/s/602955/neural-network-learns-to-identify-criminals-by-their-faces/

New AI can guess whether you're gay or straight from a photograph

An algorithm deduced the sexuality of people on a dating site with up to 91% accuracy, raising tricky ethical questions



An illustrated depiction of facial analysis technology similar to that used in the experiment. Illustration: Alamy

Credits: The Guardian. Source: https://www.theguardian.com/technology/2017/sep/07/new-artificial-intelligence-can-tell-whether-youre-gay-or-straight-from-a-photograph

How Target Figured Out A Teen Girl Was Pregnant Before Her Father Did

Kashmir Hill, FORBES STAFF
Welcome to The Not-So Private Parts where technology & privacy collide FULL BIO

Every time you go shopping, you share intimate details about your consumption patterns with retailers. And many of those retailers are studying those details to figure out what you like, what you need, and which coupons are most likely to make you happy. Target TGT -0.72%, for example, has figured out how to data-mine its way into your womb, to figure out whether you have a baby on the way long before you need to start buying diapers.

Charles Duhigg outlines in the New York Times how Target tries to hook parents-to-be at that crucial moment before they turn

TARGET

Target has got you in its aim

Credits: Forbes. Source: https://www.forbes.com/sites/kashmirhill/2012/02/16/how-target-figured-out-a-teen-girl-was-pregnant-before-her-father-did/#57a1c7576668

THE Sun HOME | FOOTBALL | SPORT | TV & SHOWBIZ | NEWS | FABULOUS | MONEY | MOTORS | TRAVEL | TECH | DEAR DEIDR

THOUSANDS of Brits are telling thieves their homes are empty and prime targets after boasting about their holidays on social media.

Nearly 700,000 people share information about their holiday plans online a month, with experts warning that tech-savvy burglars are using people's desire to "keep up with the Joneses" to know what to target, and when.

Credits: The Sun. *Thousands of Brits desperate to show off to their friends are putting their homes at risk of being BURGLED*. Source: https://www.thesun.co.uk/news/4119826/thousands-of-brits-desperate-to-show-off-to-their-friends-are-putting-their-homes-at-risk-of-being-burgled/

Facebook

Facebook sorry – almost – for secret psychological experiment on users

- Emotion-altering project should have been 'done differently'
- Company was 'unprepared' for negative reactions

This article is 3 years old

3531 109

Dominic Rushe in New York

@dominicru

Thursday 2 October 2014 21.54 BST

Facebook tampered with the news feeds of 700,000 users in an effort to alter users' emotional state. Photograph: Dado Ruvic/Reuters

Credits: The Guardian. Source: https://www.theguardian.com/technology/2014/oct/02/facebook-sorry-secret-psychological-experiment-users

The Future of Free Speech, Trolls, Anonymity and Fake News Online

Many experts fear uncivil and manipulative behaviors on the internet will persist – and may get worse. This will lead to a splintering of social media into AI-patrolled and regulated 'safe spaces' separated from free-for-all zones. Some worry this will hurt the open exchange of ideas and compromise privacy

BY LEE RAINIE, JANNA ANDERSON AND JONATHAN ALBRIGHT

Credits: Pew Internet. Source: http://www.pewinternet.org/2017/03/29/the-future-of-free-speech-trolls-anonymity-and-fake-news-online/

MORAL MACHINE

Home Judge Classic Design Browse About Feedback En

What should the self-driving car do?

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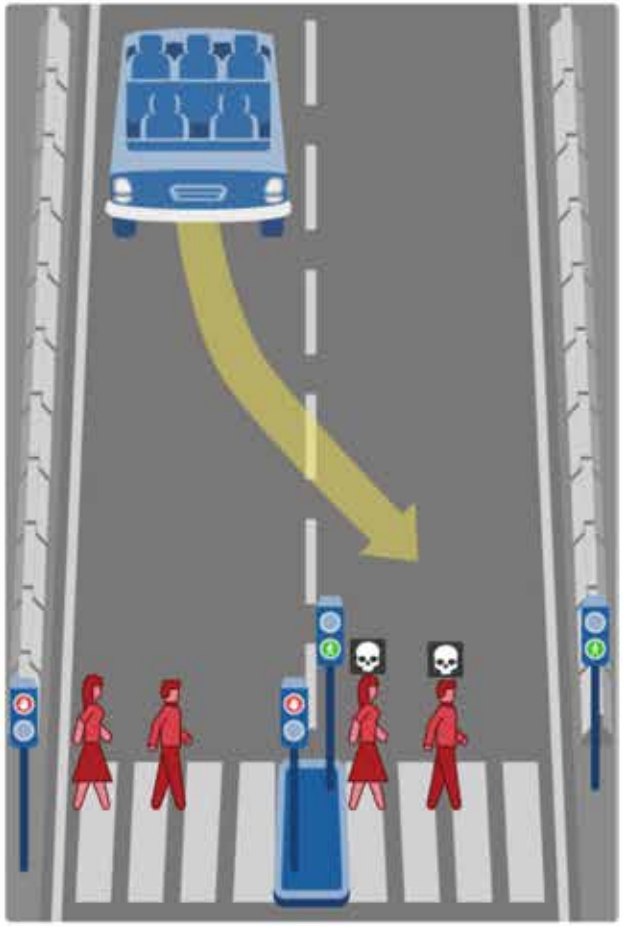

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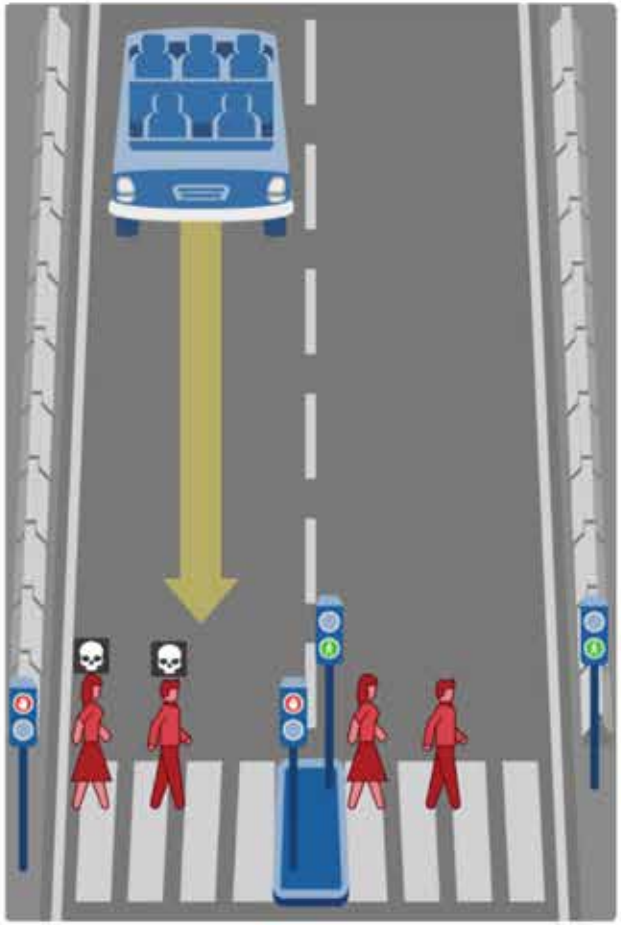

Show Description

Ezekiel Edwards, Contributor
Director of the ACLU Criminal Law Reform Project

Predictive Policing Software Is More Accurate At Predicting Policing Than Predicting Crime

Predictive policing is an unproven tool. Its current flaws would result in greater harm than help.

08/31/2016 02:58 pm ET

ARTOLYMPIC VIA GETTY IMAGES

Credits: *Huffington Post*. Source: https://www.huffingtonpost.com/entry/predictive-policing-reform_us_57c6ffe0e4b0e60d31dc9120

Ethics Exercise

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Moral and Ethical Analyses

Data gathering:

Poor data gathering methods can lead to…

Data bias:

Underlying (incomplete) data admits model bias.

TARGET

Privacy by association:

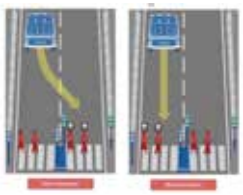
Individual's data can adversely affect self and others.

Platform leverage:

Systems can use data for unscrupulous ends.

Confirmation bias:

A feedback loop on our behaviors. Can be manipulated by others.



Trolley problem:

Systems need moral values built in.

Societal use:

System use can lend an empirical justification for problematic use.

Last Activity – A code of ethics

Pick one of the issues illustrated, linking it back to machine learning OR your own project work.

Illustrate a dilemma that you might encounter that takes the form of the issue.

Propose rule(s) that you think a data scientist should follow when encountering the dilemma involving the issue.

You may find the rest of the slides in this section useful as a reference. Relate them back to your scenario.

IS 1103: Computing and Society

Principle of Utility: An action is right to the extent that it increases the total happiness of the affected parties.

Kantianism: Act only from moral rules that you can universalize for everyone else too.

Subjective Relativism: What's right for you may not be right for me.

Code of Conduct

(n) A data scientist shall use reasonable diligence to detect, recognize, disclose and factor real, perceived and potentially hidden risks in using data science. The prudent data scientist understands that data creators and the designers and builders of data management systems have more knowledge than the data scientist and can hide risks in the foundations and interpretations / bias of the raw, created and manipulated data. The data scientist shall take reasonable remedial measures, including disclosure of risks to the client.

(o) A data scientist shall use the data science method which consists of the following steps:

(1) Careful observations of data, data sets and relationships between data;

(2) Deduction of meaning from the data and different data relationships;

(3) Formation of hypotheses;

(4) Experimental or observational testing of the validity of the hypotheses. To be termed scientific, a method of inquiry must be based on empirical and measurable evidence subject to specific principles of reasoning.

Credits: Data Science Association.
Source:
http://www.datascienceassn.org/code-of-conduct.html

Code of Conduct

(h) A data scientist shall use reasonable diligence when designing, creating and implementing machine learning systems to avoid harm. The data scientist shall disclose to the client any real, perceived or hidden risks from using a machine learning system. After full disclosure, the client is responsible for making the decision to use or not use the machine learning system. If a data scientist reasonably believes the machine learning system will cause harm, the data scientist shall take reasonable remedial measures, including disclosure to the client, and including, if necessary, disclosure to the proper authorities. The data scientist shall take reasonable measures to persuade the client to use the machine learning system appropriately.

Credits: Data Science Association.
Source:
http://www.datascienceassn.org/code-of-conduct.html

Code of Conduct

(j) A data scientist shall not engage in "Cherry picking" (pointing to individual cases or data that seem to confirm a particular position, while ignoring a significant portion of related cases or data that may contradict that position) when conducting data science. The data scientist understands that engaging in "Cherry picking" may constitute scientific fraud, suppressing evidence, or the fallacy of incomplete evidence.

(k) A data scientist shall not present incomplete evidence as real data science evidence. A data scientist may present a theory constituting incomplete evidence but shall label and clearly communicate the use of incomplete evidence.

Credits: Data Science Association.
Source:
http://www.datascienceassn.org/code-of-conduct.html

Sharing Data

- Transparency around the collection of data and how this collected data will be used;
- Clear provenance of data so that data scientists are always aware of where their datasets come from;
- Equitable and ethical access to data once it's collected, so that you don't just have a few large companies buying up data and gaining an advantage;
- Ensuring the quality of collected data; and holding sources accountable for low-quality or actively misleading data.

POLICY & ETHICS

Will Democracy Survive Big Data and Artificial Intelligence?

We are in the middle of a technological upheaval that will transform the way society is organized. We must make the right decisions now

By Dirk Helbing, Bruno S. Frey, Gerd Gigerenzer, Ernst Hafen, Michael Hagner, Yvonne Hofstetter, Jeroen van den Hoven, Roberto V. Zicari, Andrej Zwitter on February 25, 2017

Credits: Pew Internet. Source: https://www.scientificamerican.com/article/will-democracy-survive-big-data-and-artificial-intelligence/

Value Sensitive Design

Stakeholders

Drawing from Value Sensitive Design methodology, the Stakeholder criterion emphasizes the range of effects of a technology, both on those who are in direct contact with a technology (direct stakeholders), and on those who might not be direct users, but whose lives are nevertheless affected by various interactions around the technology (indirect stakeholders).

Time

Inspired by the long-term perspective of urban planning, the Time criterion helps guide designers to consider the longer term implications of their work – implications that will only emerge after the technology has moved through initial phases of novelty to later phases of appropriation and integration into society.

Values

The Value criterion emphasizes the impact of technology on human values. Our use of the term values draws from the Value Sensitive Design literature, "what a person or group of people consider important in life." In interaction design, we have found values of interest to include but are not limited to: autonomy, community, cooperation, democratization, environmental sustainability, expression, fairness, human dignity, inclusivity and exclusivity, informed consent, justice, ownership, privacy, self-efficacy, security, trust, and universal access.

Pervasiveness

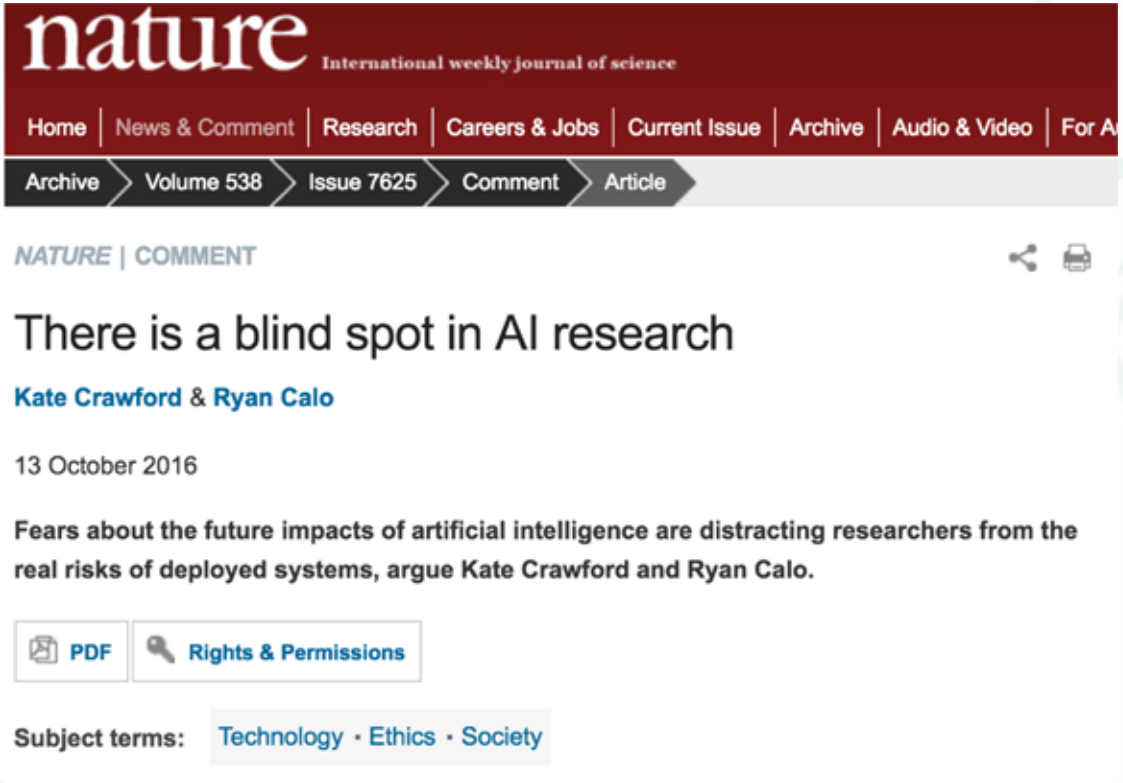
The Pervasiveness criterion emphasizes systemic interactions that follow from the widespread adoption of an interactive technology. Technologies can become pervasive with respect to geographic (e.g., city navigation software use within urban areas), cultural (e.g., text messaging within the deaf community), demographic (e.g., online social networking sites among teenagers), and other factors.

Credits: Value Sensitive Design. Univ. of Washington. *The Envisioning Cards.* http://www.envisioningcards.com/

The blind spot in AI research

We must take responsibility for the safeguards of the technology we create.

You will be tested in your work environment in years to come. Remember to think through its consequences.



nature International weekly journal of science

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Archive > Volume 538 > Issue 7625 > Comment > Article

NATURE | COMMENT

There is a blind spot in AI research

Kate Crawford & Ryan Calo

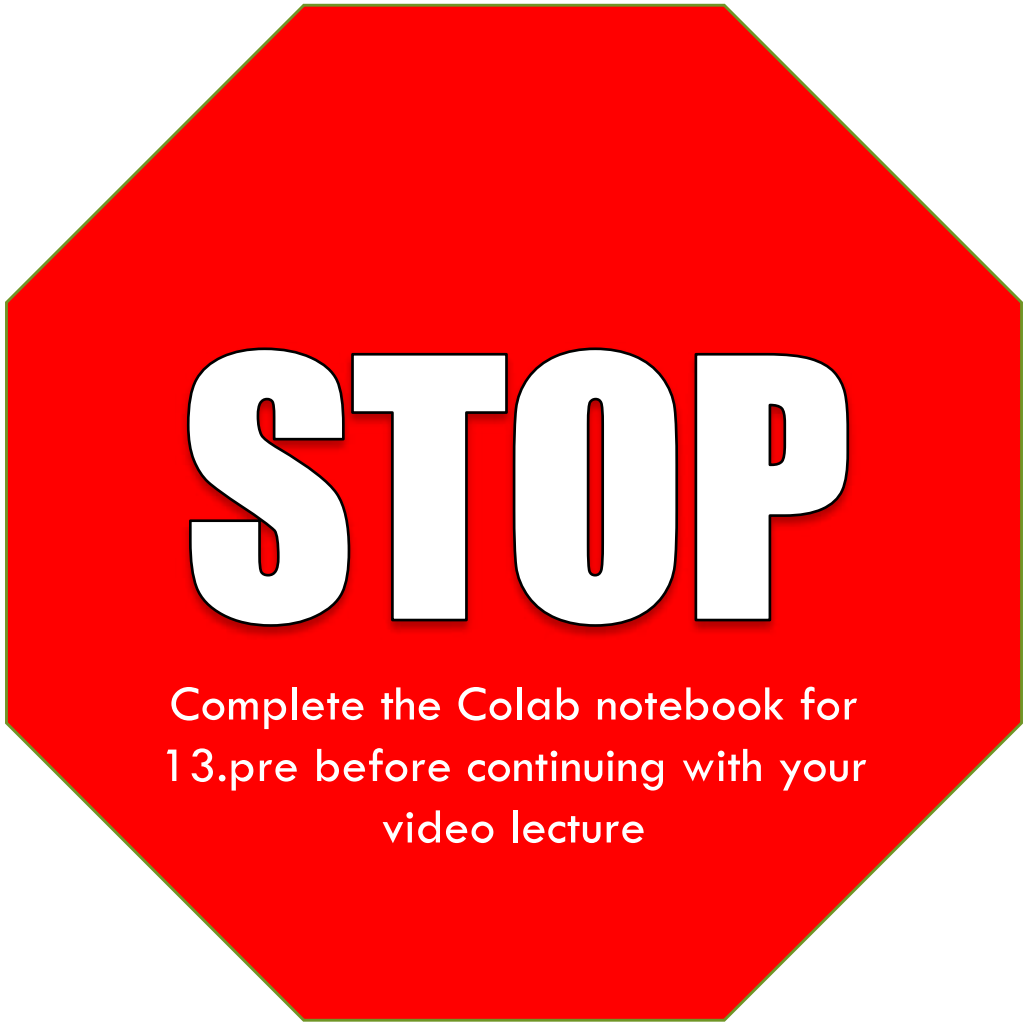
13 October 2016

Fears about the future impacts of artificial intelligence are distracting researchers from the real risks of deployed systems, argue Kate Crawford and Ryan Calo.

PDF | Rights & Permissions

Subject terms: Technology · Ethics · Society

Credits: *Nature*. http://www.nature.com/news/there-is-a-blind-spot-in-ai-research-1.20805


STOP
Complete the Colab notebook for 13.pre before continuing with your video lecture

Bias

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What do you see?

- Bananas
- Bananas at a store
- Bananas on shelves
- Bunches of bananas
- Bananas with stickers on them

Credits: *Huffington Post*. Source: https://www.huffingtonpost.com/entry/predictive-policing-reform-us_57e6ffe0e4b0e60d31dc9120

Credits: Mitchell et al. (2017) *Algorithmic Bias in Artificial Intelligence: The Seen and Unseen Factors influencing machine perception of images and language*. Used with permission.

What do you see?

- Green Bananas
- Unripe Bananas

Credits: Mitchell et al. (2017) *Algorithmic Bias in Artificial Intelligence: The Seen and Unseen Factors influencing machine perception of images and language*. Used with permission.

What do you see?

- Overripe bananas
- Bananas with spots
- Bananas good for baking banana bread

Credits: Mitchell et al. (2017) *Algorithmic Bias in Artificial Intelligence: The Seen and Unseen Factors influencing machine perception of images and language*. Used with permission.

What do you see?

- Yellow Bananas

- Yellow is prototypical for bananas

Credits: Mitchell et al. (2017) *Algorithmic Bias in Artificial Intelligence: The Seen and Unseen Factors influencing machine perception of images and language*. Used with permission.

How could this be?

A man and his son are in a terrible accident and are rushed to the hospital in critical care.

The doctor looks at the boy and exclaims "I can't operate on this boy, he's my son!”

How could this be?

Credits: Mitchell et al. (2017) *Algorithmic Bias in Artificial Intelligence: The Seen and Unseen Factors influencing machine perception of images and language*. Used with permission.

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The majority of test subjects overlooked the possibility that the doctor is a she - including men, women, and self-described feminists.
Wapman & Belle, Boston University

Credits: Mitchell et al. (2017) *Algorithmic Bias in Artificial Intelligence: The Seen and Unseen Factors influencing machine perception of images and language*. Used with permission.

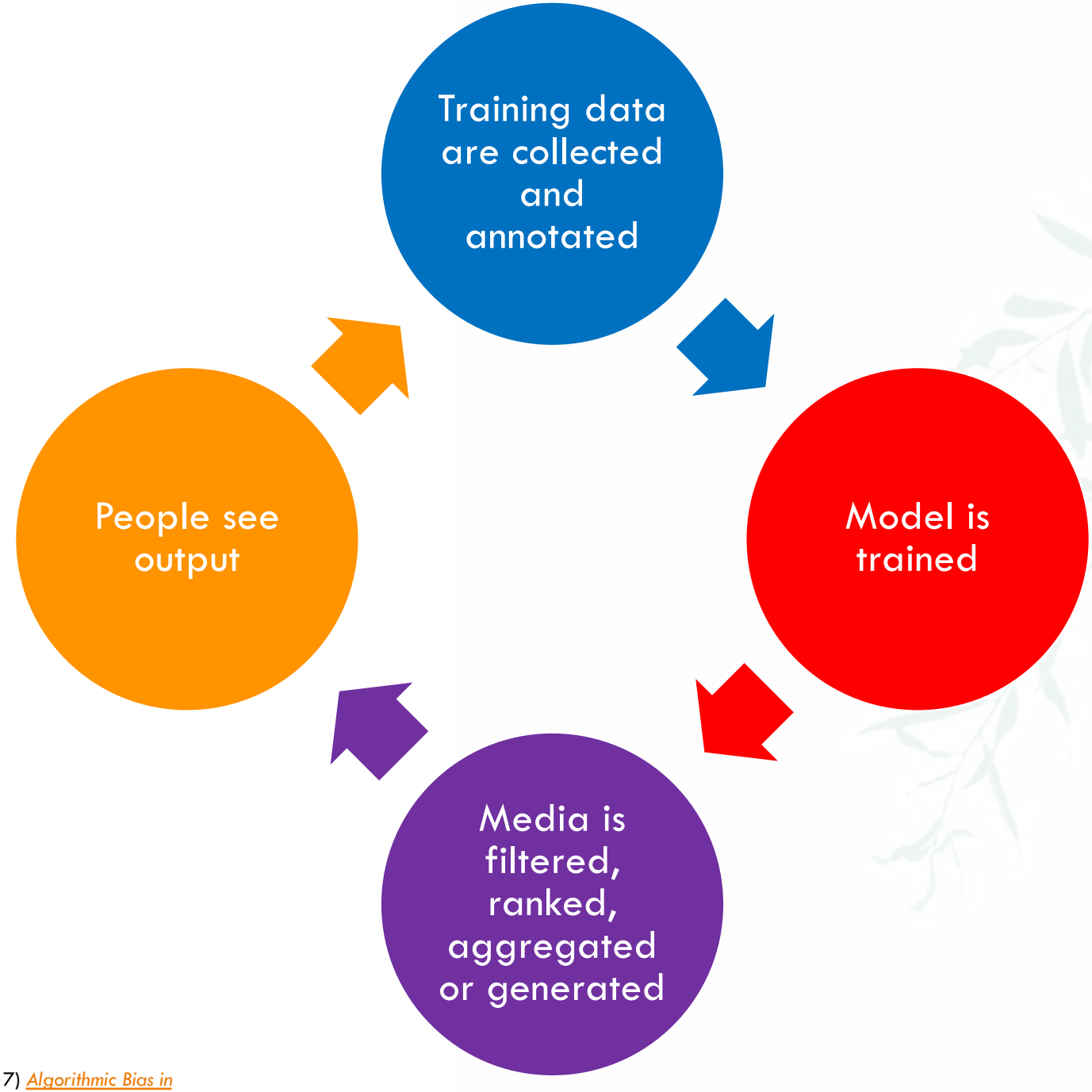
World learning from text

If aliens remotely studied our news reports, what would they learn about us?

Word	Frequency in corpus
"spoke"	11,577,917
"laughed"	3,904,519
"murdered"	2,834,529
"inhaled"	984,613
"breathed"	725,034
"hugged"	610,040
"blinked"	390,692
"exhale"	168,985

(Data from Gordon and Van Durme, 2013)

Credits: Mitchell et al. (2017) *Algorithmic Bias in Artificial Intelligence: The Seen and Unseen Factors influencing machine perception of images and language*. Used with permission.



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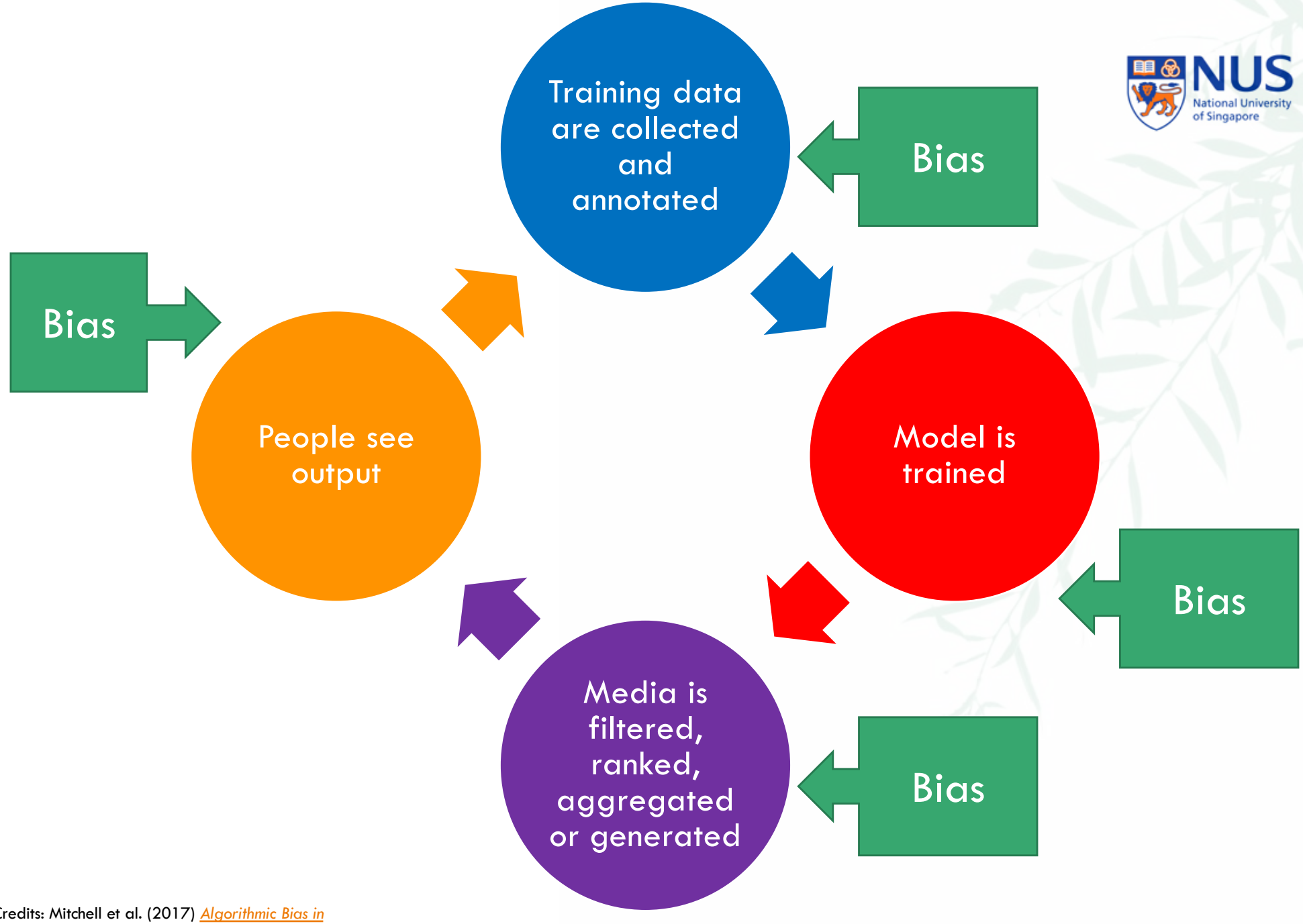

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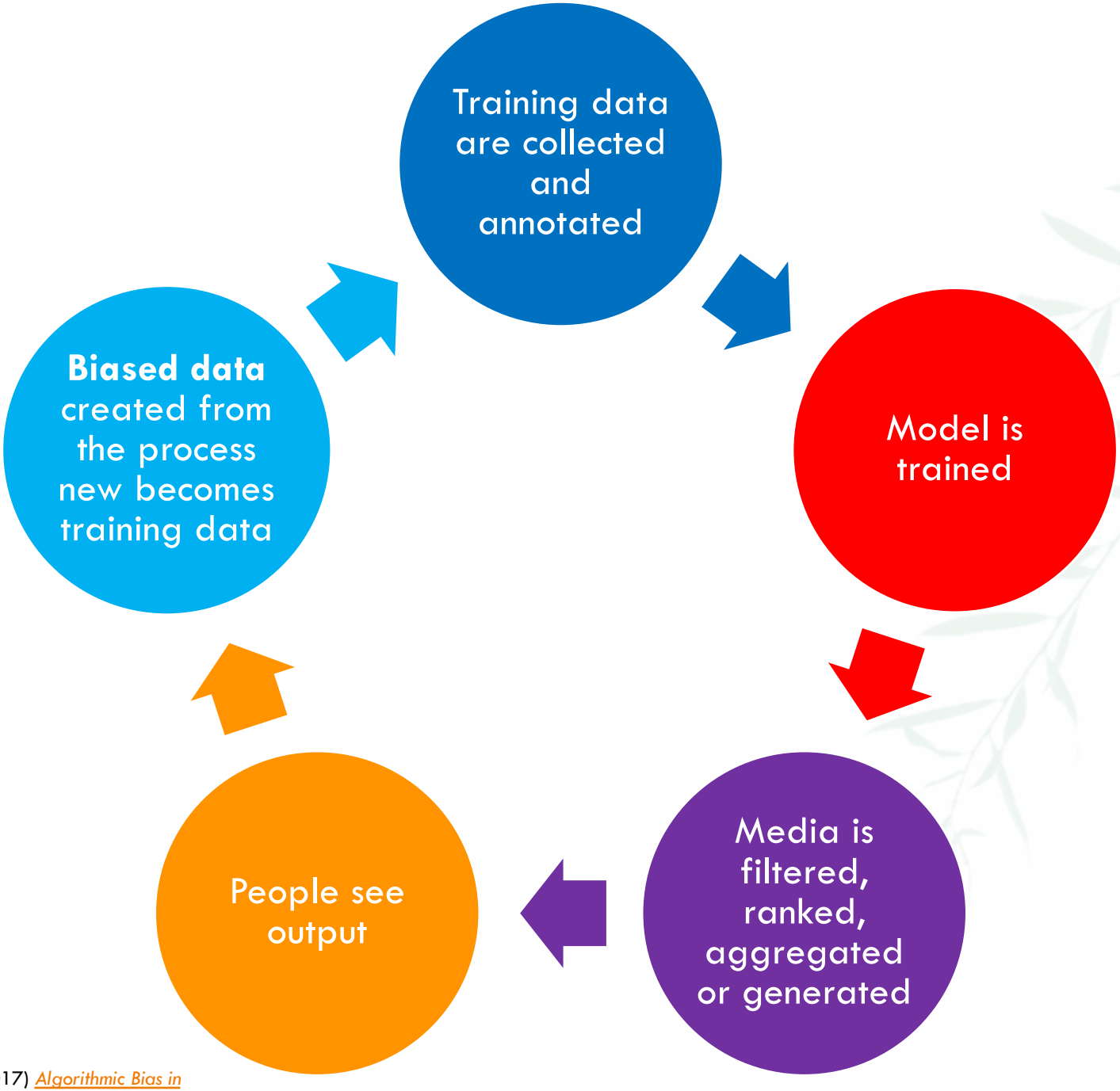

Credits: Mitchell et al. (2017) *Algorithmic Bias in Artificial Intelligence: The Seen and Unseen Factors influencing machine perception of images and language*. Used with permission.

Types of Bias

https://developers.google.com/machine-learning/crash-course/fairness/types-of-bias

Reporting Bias

Automation Bias

Selection Bias

Group Attribution Bias

Implicit Bias

and many more…

Fairness: Evaluating for Bias

Metrics calculated against an entire set don't always give an accurate picture of how fair the model is.

A model developed to predict the presence of tumors that is evaluated against a validation set of 1,000 records.

Q: Anything wrong with using this for both male and female diagnosis?

True Positives (TPs): 16	False Positives (FPs): 4
False Negatives (FNs): 6	True Negatives (TNs): 974

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{16}{16 + 4} = 0.800$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{16}{16 + 6} = 0.727$$

Credits: Google Crash Course in Machine Learning
Evaluating for Fairness

Fairness: Evaluating for Bias

Metrics calculated against an entire set don't always give an accurate picture of how fair the model is.

A model developed to predict the presence of tumors that is evaluated against a validation set of 1,000 records.

500 records are from female patients,

500 records are from male patients.

True Positives (TPs): 16	False Positives (FPs): 4
False Negatives (FNs): 6	True Negatives (TNs): 974

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{16}{16 + 4} = 0.800$$

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Credits: Google Crash Course in Machine Learning
Evaluating for Fairness

Fairness: Evaluating for Bias

Metrics calculated against an entire set don't always give an accurate picture of how fair the model is.

A model developed to predict the presence of tumors that is evaluated against a validation set of 1,000 records. 500 records are from females, 500 from males.

Female Patient Results

True Positives (TPs): 10	False Positives (FPs): 1
False Negatives (FNs): 1	True Negatives (TNs): 488

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{10}{10 + 1} = 0.909$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{10}{10 + 1} = 0.909$$

Male Patient Results

True Positives (TPs): 6	False Positives (FPs): 3
False Negatives (FNs): 5	True Negatives (TNs): 486

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{6}{6 + 3} = 0.667$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{6}{6 + 5} = 0.545$$

Credits: Google Crash Course in Machine Learning
Evaluating for Fairness